

OR

Q - 2 Answer the following questions.

- (A) To draw a circle on the output screen whose parameters are user defined? [04]
- (B) Explain type casting with example. [06]
- (C) What is constructor? Why we use it? Can we overload the constructor? If yes then How? [04]

Q - 3 Answer the following questions.

- (A) What is friend function? Explain it with simple example. [04]
- (B) Draw a control structure for the following syntax. [06]

```
if (condition 1)
{
    if (condition 2)
    {    action 1;    }
    else
    {    action 2;    }
} else
{    action 3;    }
```

- (C) What is Virtual Base Classes? Why it is needed? Explain it with Example [04]

OR

Q - 3 Answer the following questions.

- (A) What is Copy Constructor? Explain with example [04]
- (B) What is the difference between Call by Reference and Call by value? Explain it with swapping Example. [06]
- (C) What is Type Conversion? State the difference between Implicit Conversion and Explicit Conversion. List out the type of data conversion. [04]

SECTION - II

Q - 4 Answer the following questions.

- 1) State true or false: private member of base class can be derived in derive class.
- 2) Unary operators, overloaded by means of a member function, take _____ arguments and overloaded by means of a friend function, take _____ arguments.
- 3) Binary operators, overloaded by means of a member function, take _____ arguments and overloaded by means of a friend function, take _____ arguments.
- 4) Which is not derived data type?
 - a) Array
 - b) Function
 - c) Pointer
 - d) Integer

- 5) State true or false: A destructor takes any argument and returns any value.
- 6) State true or false: An abstract class is not used to create objects.
- 7) B is a class. bptr is pointer of type B and b is an object of class B. D is derived from class B. dptr is pointer of type D and d is an object of class D. which one is incorrect representation from the following.
 - a) bptr = &b
 - b) dptr = &b
 - c) bptr = &d
 - d) dptr = &d

Q - 5 Answer the following questions.

(A) Define a class employee with following specifications:

[04]

Private members of class employee:

empno – integer

ename – size = 20 – character

basic, hra, da – float

netpay – float

Calculate () a function to find basic + hra + da with float return type.

Public member functions of class employee:

Have data() function to accept value for empno, ename, basic, hra, da and invoke calculate() to calculate netpay

dispdata () function to display all the members on the screen.

(B) Attempt any three.

[03]

1) _____ diagram is used to show the relationship between objects & Attributes.

2) State True or False: "Operation is data value held by the objects in the class."

3) _____ describes a group of Objects with similar properties and Common behavior.

4) List out the components of Sequence Diagram.

(C) Draw class diagram for Railway Reservation system.

[07]

OR

Q - 5 Answer the following questions.

(A) Write a program to generate the following series:

[04]

1

12

123

1234

12345

(B) Explain the step of Object Modeling technique (OMT).

[03]

(C) Draw sequence diagram for ATM.

[07]

Q - 6 Answer the following questions.

(A) Draw use case diagram for Library Management system.

[05]

(B) Describe the importance of design by contract in model.

[03]

(C) List out and Explain purposes of Modeling.

[02]

(D) Explain Multiplicity with suitable example.

[02]

(E) What is the output of the following?

[02]

Ans = $(5 * (4/2) + (3/(3+8))) - 2$

OR

Q - 6 Answer the following questions.

(A) Draw activity diagram for Order placing system.

[05]

(B) What is Event, Transition and State in terms of state chart diagram?

[03]

(C) List out components of activity diagram and explain with example.

[02]

(D) What do you mean by aggregation and composition?

[02]

(E) What is the output of the following?

[02]

If we take $a = 3$ and $b = 2$ find 1) $a \& b$ 2) $a | b$
